

IMAT MOCK EXAM 6 — "At-Level Calibration"

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 45/90.0 · Mock exam · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Logical Reasoning & Reading (Q1-9)

1. "Bacteria store fragments of viral DNA in ordered arrays (CRISPR loci) in their own genome. These fragments are transcribed into guide molecules used by Cas proteins, which cut any invading DNA whose sequence matches a guide. A virus escapes this defence only if it mutates the exact targeted sequence. Because the array is inherited, daughter cells of a survivor remain protected."

Which of the following is not supported by the passage?

- A)** A bacterium that survives infection can pass protection on to its offspring.
- B)** The defence is guided by RNA copies of stored viral sequences.
- C)** Mutation of the targeted sequence can let a virus evade the defence.
- D)** During infection, bacteria deliberately rewrite their own DNA to match the invading virus.
- E)** The system behaves as a form of inherited immunity.

Answer: D

The passage describes storing fragments of viral DNA and transcribing them into guide molecules; it never says the bacterium rewrites its own DNA during infection to match the invader, which would be the opposite of passive fragment storage. Protection passing to offspring, RNA guides, escape via mutation of the targeted sequence, and the inherited-immunity behaviour are each stated or directly implied in the passage.

2. Five friends P, Q, R, S and T sit in a row of five seats, all facing the same direction, with seat 1 at the left end and seat 5 at the right end. Q sits immediately to the right of P. R does not sit adjacent to P. S sits at an end seat. T sits somewhere to the left of S.

Which of the following must be true?

- A)** R sits between P and T.
- B)** P sits at the far left.
- C)** Q is not adjacent to R.
- D)** T is not at an end seat.
- E)** S sits at seat 5 (the far right).

Answer: E

Givens: seats 1 to 5 left to right; Q immediately right of P; R not adjacent to P; S at an end seat (1 or 5); T somewhere left of S.

Deduction 1: T needs a seat to its left, so S cannot sit at seat 1, because no seat lies to the left of seat 1.

Deduction 2: S must sit at an end, and seat 1 is impossible, so S sits at seat 5 in every valid arrangement.

Check: the layout T in 1, P in 2, Q in 3, R in 4, S in 5 satisfies every given (Q is immediately right of P, R is not adjacent to P, S is at the right end, T is left of S).

Conclusion: S sits at seat 5, the far right, must be true.

That same layout kills the tempting alternatives: P is not at the far left, T does occupy an end seat, R is not between P and T, and Q does sit adjacent to R.

3. (General knowledge) The 2026 Winter Olympic Games, held in February 2026, are hosted by:

- A) Milano and Cortina d'Ampezzo, Italy
- B) Venice, Italy
- C) Turin, Italy
- D) Palermo, Italy
- E) Rome, Italy

Answer: A

The 2026 Winter Olympic Games (6 to 22 February 2026) are hosted by Milano and Cortina d'Ampezzo in Italy, the country's third Winter Games after Cortina d'Ampezzo 1956 and Turin 2006. Turin is the tempting recall, but it hosted in 2006, not 2026.

4. No reptile is warm-blooded. Some reptiles are kept as pets.

Which conclusion necessarily follows?

- A) No pet is a reptile.
- B) Some pets are warm-blooded.
- C) Some pets are not warm-blooded.
- D) Some warm-blooded animals are kept as pets.
- E) All pets are cold-blooded.

Answer: C

Givens: (1) no reptile is warm-blooded; (2) some reptiles are kept as pets.

Deduction 1: from (2), at least one animal is both a reptile and a pet.

Deduction 2: by (1), that animal, being a reptile, is not warm-blooded.

Deduction 3: therefore at least one pet is not warm-blooded.

Conclusion: some pets are not warm-blooded follows necessarily.

The tempting claim that all pets are cold-blooded overshoots: the premises say nothing about pets that are not reptiles, so warm-blooded non-reptile pets remain possible.

5. A clock shows 4:40. What is the angle between the hour hand and the minute hand?

- A) 40°
- B) 60°
- C) 100°
- D) 120°
- E) 260°

Answer: C

Given: time 4:40; the minute hand sweeps 6° per minute; the hour hand sweeps 30° per hour plus 0.5° per minute.

Relationship: minute hand angle from 12 = minutes $\times 6^\circ$; hour hand angle from 12 = hours $\times 30^\circ$ + minutes $\times 0.5^\circ$.

Substitute: minute hand = $40 \times 6^\circ = 240^\circ$

Substitute: hour hand = $4 \times 30^\circ + 40 \times 0.5^\circ = 120^\circ + 20^\circ = 140^\circ$

Angle between the hands = $240^\circ - 140^\circ = 100^\circ$

Result: 100° .

Classic slip: leaving the hour hand frozen at exactly 4 o'clock gives $240^\circ - 120^\circ = 120^\circ$, ignoring the extra 20° it creeps forward in 40 minutes.

6. An enzyme was assayed at three pH values. Without inhibitor, the measured rates at pH 5, 7 and 9 were 20, 100 and 20 respectively. With a fixed concentration of inhibitor I present, the rates were 18, 50 and 18 at the same three pH values.

Which conclusion is best supported by the data?

- A) The enzyme is inactive at pH 9.
- B) The enzyme is most active at pH 7, and the inhibitor's relative effect is also greatest at pH 7.
- C) The enzyme's optimum lies between pH 7 and pH 9.
- D) The inhibitor reduces the rate by the same fraction at every pH.
- E) The inhibitor destroys the enzyme at acidic pH.

Answer: B

Givens: rates without inhibitor 20 (pH 5), 100 (pH 7), 20 (pH 9); with inhibitor 18, 50, 18 at the same pH values.

Deduction 1: the uninhibited maximum is 100 at pH 7, so the optimum is at pH 7 itself, not between 7 and 9.

Deduction 2: relative loss at pH 5 = $(20 - 18) \div 20 = 2 \div 20 = 0.10$, a 10% reduction

Deduction 3: relative loss at pH 7 = $(100 - 50) \div 100 = 50 \div 100 = 0.50$, a 50% reduction

Deduction 4: relative loss at pH 9 = $(20 - 18) \div 20 = 0.10$, a 10% reduction

Deduction 5: the fractional effect differs with pH and is greatest at pH 7, so inhibition is not a constant fraction.

Result: the enzyme is most active at pH 7 and the inhibitor's relative effect is also greatest at pH 7.

The claim that the enzyme is inactive at pH 9 fails because a rate of 20 is still measured there, and the residual 18 of 20 at pH 5 shows no destruction at acidic pH.

7. A large national audit reports that patients operated on in city hospitals have fewer post-surgical complications than patients operated on in small rural hospitals. The authors conclude that the quality of surgical care is worse in rural hospitals.

Which of the following, if true, most weakens the authors' conclusion?

- A) Rural hospitals carry out a higher proportion of emergency operations, which have higher complication rates in any setting.
- B) Rural surgeons work longer shifts than city surgeons.
- C) Some city hospitals are also teaching hospitals.
- D) City hospitals have more beds than rural hospitals.
- E) City hospitals have newer pieces of equipment on average.

Answer: A

The argument treats the complication rate as a pure measure of surgical quality. If rural hospitals perform a higher share of emergency operations, which carry higher complication rates in any setting, the observed gap could reflect how severe the cases are rather than how good the care is, cutting the link between the evidence and the conclusion. Longer rural shifts and newer city equipment would if anything strengthen the conclusion, and bed counts or teaching status say nothing about case severity.

8. A sequence begins 2, 4, 7, 14, 17, 34, ... following a repeating two-step rule. What is the next term?

- A) 35 B) 37 C) 51 D) 68 E) 71

Answer: B

Given: the terms 2, 4, 7, 14, 17, 34 follow a repeating two-step rule.

Test the pattern: $2 \times 2 = 4$ (double), $4 + 3 = 7$ (add 3), $7 \times 2 = 14$ (double), $14 + 3 = 17$ (add 3), $17 \times 2 = 34$ (double).

The rule alternates double then add 3, and it reproduces every given term.

Since 34 was reached by doubling, the next step is add 3.

$$34 + 3 = 37$$

Result: 37.

Classic slip: doubling again to 68 keeps multiplying instead of switching back to the +3 step of the alternating rule.

9. A jacket costing 120 is discounted by 20%, and the new price is then discounted by a further 25%. What is the total percentage reduction from the original price?

- A) 35% B) 40% C) 42.5% D) 45% E) 46%

Answer: B

Given: original price 120; discounts of 20% then 25%, each taken on the current price.

Relationship: successive discounts multiply; each step multiplies the running price by $(1 - \text{discount})$.

$$\text{Substitute step 1: } 120 \times (1 - 0.20) = 120 \times 0.80 = 96$$

$$\text{Substitute step 2: } 96 \times (1 - 0.25) = 96 \times 0.75 = 72$$

$$\text{Total reduction} = 120 - 72 = 48$$

$$\text{Percentage reduction} = 48 \div 120 = 0.40 = 40\%$$

Result: 40%.

Classic slip: adding 20% + 25% to get 45% wrongly treats the second discount as taken on the original 120 instead of on the reduced 96.

Section B - Biology (Q10-32)

10. In a dihybrid cross $AaBb \times AaBb$ with two unlinked genes (complete dominance at both), what is the probability that an offspring has genotype $AaBb$?

- A) 1/16 B) 1/4 C) 1/2 D) 3/16 E) 9/16

Answer: B

From $Aa \times Aa$ the probability of genotype Aa is $1/2$, and from $Bb \times Bb$ the probability of Bb is $1/2$; the genes are unlinked, so multiply: $1/2 \times 1/2 = 1/4$. Nine-sixteenths instead gives the chance of showing both dominant phenotypes, a different question from the double heterozygote genotype.

11. Which cross is expected to produce a phenotypic ratio of 9:3:3:1 (two traits, complete dominance)?

- A) $Aa \times Aa$ B) $AABB \times aabb$ C) $AABb \times AaBB$ D) $AaBb \times AaBb$ E) $AaBb \times aabb$

Answer: D

Only the double heterozygote cross $AaBb \times AaBb$ yields 9 both-dominant : 3 first-trait-only : 3 second-trait-only : 1 double recessive, since both genes segregate 3:1 independently. Crossing a double heterozygote with a double recessive is the testcross giving 1:1:1:1, the monohybrid $Aa \times Aa$ gives only 3:1, and crossing homozygous parents gives uniform offspring.

12. A mother is a carrier of red-green colour blindness ($X^H X^h$) and the father has normal vision ($X^H Y$). What fraction of their sons is expected to be colour-blind?

- A)** 3/4 **B)** 1/4 **C)** all of them **D)** 0 **E)** 1/2

Answer: E

A son's single X chromosome always comes from the mother, because the father supplies the Y. The carrier mother passes X^H or X^h with equal probability, so exactly half of her sons inherit X^h and are colour-blind. One quarter would be the rate among all children rather than among sons, and the father's normal X passes only to daughters, so he contributes nothing to his sons' vision.

13. DNA replication is described as "semiconservative" because each daughter double helix:

- A)** conserves half of the parent's genes
B) conserves only the sugar-phosphate backbone of the parent
C) is made entirely of new nucleotides
D) contains randomly mixed old and new nucleotides within each strand
E) contains one intact parental strand and one newly synthesised strand

Answer: E

Each daughter double helix keeps one intact parental strand paired with one newly synthesised strand, because each parental strand templates its partner; Meselson and Stahl confirmed exactly this result. Molecules made entirely of new nucleotides describe the conservative model, and old and new segments randomly mixed within single strands describe the dispersive model, both of which that experiment excluded.

14. The enzyme that synthesises an RNA strand from a DNA template during transcription is:

- A)** helicase
B) DNA polymerase
C) reverse transcriptase
D) RNA polymerase
E) primase

Answer: D

RNA polymerase reads the DNA template 3' to 5' and joins ribonucleotides into an RNA chain 5' to 3'; that is the defining enzyme of transcription. Reverse transcriptase copies RNA into DNA, DNA polymerase copies DNA into DNA, and helicase only unwinds the parental double helix.

15. Which nucleic-acid molecule physically carries the anticodon?

- A)** small nuclear RNA
B) mRNA
C) tRNA
D) rRNA
E) the DNA template strand

Answer: C

The anticodon is the three-base sequence on the anticodon loop of transfer RNA, and it base-pairs with the complementary codon on messenger RNA during translation. Messenger RNA carries codons rather than anticodons, and ribosomal RNA provides the structure and catalysis of the ribosome rather than codon recognition.

16. The α -helix of a protein is stabilised primarily by:

- A)** hydrogen bonds between backbone groups
- B)** hydrophobic cores
- C)** ionic bonds between consecutive amino acids
- D)** peptide bonds between side chains
- E)** disulfide bridges

Answer: A

The α -helix is secondary structure held together by hydrogen bonds between backbone C=O and N-H groups, giving the regular coiled pattern. Disulfide bridges belong to tertiary structure, and the side chains project outward from the helix without taking part in its stabilising pattern.

17. A point mutation changes an mRNA codon from AUG (methionine) to UGA. UGA is a stop codon. The most likely consequence is:

- A)** no change in the protein
- B)** a frameshift downstream
- C)** substitution of one amino acid by another
- D)** premature termination, giving a truncated protein
- E)** an elongated protein

Answer: D

Changing the methionine codon into the stop codon UGA turns a sense codon into a stop signal, a nonsense mutation: the ribosome releases the chain early and the protein is truncated. Only one codon is altered, so the downstream reading frame is intact and no frameshift occurs, and premature termination shortens rather than elongates the product.

18. In aerobic respiration, CO₂ is released:

- A)** on the inner mitochondrial membrane, during electron transport
- B)** in the cytosol, during glycolysis
- C)** in the mitochondrial matrix, during pyruvate oxidation and the Krebs cycle
- D)** in the intermembrane space, during proton pumping
- E)** by ATP synthase

Answer: C

All six CO₂ per glucose are released in the mitochondrial matrix: two during pyruvate oxidation (pyruvate to acetyl-CoA) and four across the two turns of the Krebs cycle, at the decarboxylations of isocitrate and α -ketoglutarate. Glycolysis runs in the cytosol and releases no CO₂, and the electron transport chain on the inner membrane transfers electrons to oxygen without any decarboxylation step.

19. Cyanide blocks complex IV of the electron transport chain. Why does ATP synthesis by ATP synthase then stop?

- A)** Electron transfer stops, protons are no longer pumped, and the gradient that powers ATP synthase dissipates.
- B)** ATP synthase can run without a gradient for only a few minutes.
- C)** ATP synthase is directly inhibited by cyanide.
- D)** Glycolysis stops, so there is no pyruvate.
- E)** The Krebs cycle accelerates and floods the matrix with protons.

Answer: A

ATP synthase is driven by the proton gradient that the electron transport chain builds. Blocking complex IV stops electron flow, protons are no longer pumped, and the gradient dissipates, so there is no proton-motive force left to power the synthase. Cyanide never touches ATP synthase itself, glycolysis and pyruvate supply continue, and there is no mechanism by which the Krebs cycle would accelerate and flood the matrix with protons.

20. The main purpose of the fermentation step (lactic acid or alcoholic) in anaerobic conditions is to:

- A)** eliminate CO₂ from the cell
- B)** synthesise NADH for the electron transport chain
- C)** produce additional ATP beyond glycolysis
- D)** remove glucose from the cell
- E)** regenerate NAD⁺ so glycolysis can continue

Answer: E

Fermentation's purpose is to re-oxidise the NADH produced by glycolysis back to NAD⁺, because glycolysis halts without a fresh supply of NAD⁺. It generates no ATP itself: the entire anaerobic yield (a net 2 per glucose) comes from glycolysis, lactic acid fermentation releases no CO₂, and NADH is consumed rather than synthesised in the fermentation step.

21. Two inhibitors were characterised for an enzyme. With inhibitor I, the apparent K_m is 4× higher and V_{max} is unchanged. With inhibitor II, K_m is unchanged and V_{max} falls to a quarter. Inhibitor I is best classified as:

- A)** non-competitive
- B)** uncompetitive
- C)** an allosteric activator
- D)** competitive
- E)** an irreversible poison

Answer: D

A fourfold higher apparent K_m with V_{max} unchanged is the signature of competitive inhibition: the inhibitor competes at the active site, and a high enough substrate concentration can outcompete it, so the maximum rate survives. The other inhibitor's profile, V_{max} falling to a quarter with K_m unchanged, is non-competitive; an irreversible poison would collapse V_{max} regardless of substrate level, and an activator would not raise K_m.

22. A cell's ATP is rapidly depleted by a metabolic poison. Which transport process will fail first?

- A)** water movement through aquaporins
- B)** diffusion of oxygen through the membrane
- C)** osmosis
- D)** facilitated diffusion of glucose down its gradient
- E)** Na⁺/K⁺ pump activity

Answer: E

The sodium-potassium pump is primary active transport and consumes ATP directly, so rapid ATP depletion stops it first. Water movement through aquaporins, oxygen diffusion, osmosis and facilitated diffusion of glucose down its gradient are all passive, gradient-driven processes that need no ATP.

23. The neurotransmitter released at the neuromuscular junction of vertebrates is:

- A)** acetylcholine **B)** serotonin **C)** dopamine **D)** GABA **E)** adrenaline

Answer: A

At the vertebrate neuromuscular junction the motor neurone releases acetylcholine, which binds nicotinic receptors on the muscle fibre membrane and triggers contraction. Adrenaline, dopamine, GABA and serotonin act as transmitters at central or autonomic synapses, not at this junction.

24. The resting membrane potential of a typical vertebrate neurone is closest to:

- A) -70 mV B) -55 mV C) -30 mV D) 0 mV E) $+30$ mV

Answer: A

A typical vertebrate neurone rests near -70 mV, set mainly by potassium leak channels. The -55 mV figure is the usual firing threshold and $+30$ mV the peak of the action potential, while 0 mV or -30 mV are far too positive for a resting membrane.

25. Vitamin K is required in the liver for the synthesis of:

- A) insulin
B) albumin only
C) prothrombin and other clotting factors
D) bile salts
E) haemoglobin

Answer: C

Vitamin K is the cofactor for γ -carboxylation of glutamate residues, which the liver needs to produce functional prothrombin (factor II) and factors VII, IX and X; without it, clotting fails. It plays no role in haemoglobin synthesis, bile salts are made independently of it, insulin is pancreatic rather than hepatic, and the restriction to albumin alone is wrong because the vitamin K products are the clotting factors.

26. After 12 hours without drinking, blood plasma osmolality rises. Which endocrine response helps conserve water?

- A) aldosterone secretion increases
B) ADH (vasopressin) secretion increases
C) insulin secretion increases
D) atrial natriuretic peptide secretion increases
E) cortisol secretion stops

Answer: B

Raised plasma osmolality, sensed by hypothalamic osmoreceptors, is the primary stimulus for increased ADH (vasopressin) secretion; ADH inserts aquaporins in the collecting duct so the kidney reabsorbs and conserves water. Aldosterone chiefly conserves sodium, atrial natriuretic peptide promotes salt and water loss (the opposite of conservation), and insulin or a halt in cortisol does not address water balance here.

27. In skeletal muscle contraction, which event directly exposes the myosin-binding sites on actin?

- A) calcium binding to tropomyosin
B) sodium entering the sarcoplasmic reticulum
C) calcium binding to troponin
D) ATP binding to myosin heads
E) acetylcholine binding to the sarcolemma

Answer: C

Calcium released from the sarcoplasmic reticulum binds troponin, whose shape change pulls tropomyosin away from the myosin-binding sites on actin. Assigning the calcium binding to tropomyosin itself swaps the roles of the two regulatory proteins, ATP binding to myosin belongs to cross-bridge cycling, and acetylcholine acts on the sarcolemma well before this step.

28. Antibodies circulating in plasma are produced by:

- A)** platelets
- B)** cytotoxic T cells
- C)** T helper cells
- D)** plasma cells differentiated from B lymphocytes
- E)** macrophages

Answer: D

Activated B lymphocytes differentiate into plasma cells, and it is these plasma cells that secrete the antibodies found in plasma. Helper T cells coordinate the response and cytotoxic T cells kill infected targets, but neither secretes antibodies, and macrophages engulf pathogens and present antigen instead.

29. A human cell has just completed DNA replication (S phase). Which chromosome/chromatid count is correct?

- A)** 23 chromosomes, 23 chromatids
- B)** 46 chromosomes, 46 chromatids
- C)** 46 chromosomes, 92 chromatids
- D)** 92 chromosomes, 92 chromatids
- E)** 92 chromosomes, 46 chromatids

Answer: C

Replication doubles the DNA while every centromere remains intact, so each of the 46 chromosomes now consists of two sister chromatids: 46 chromosomes and 92 chromatids. The chromosome count would only reach 92 at anaphase, when the sister chromatids separate and each begins to count as a chromosome.

30. The walls of the pulmonary alveoli are lined by:

- A)** ciliated columnar epithelium
- B)** simple squamous epithelium
- C)** transitional epithelium
- D)** stratified squamous epithelium
- E)** simple cuboidal epithelium

Answer: B

Gas exchange demands the thinnest possible barrier, so the alveolar walls are lined by simple squamous epithelium, a single layer of flattened cells. Ciliated columnar tissue lines the conducting airways, and stratified, transitional or cuboidal linings would all be far too thick for rapid diffusion.

31. The light-independent reactions of photosynthesis (Calvin cycle) take place in the:

- A)** thylakoid lumen
- B)** cytosol
- C)** thylakoid membranes
- D)** mitochondrial matrix
- E)** stroma of the chloroplast

Answer: E

The Calvin cycle enzymes, including RuBisCO for carbon fixation, are dissolved in the stroma, the fluid surrounding the thylakoids. The thylakoid membranes and their lumen host the light-dependent reactions instead, and neither the mitochondrial matrix nor the cytosol has any part in photosynthesis.

32. Yeast cells are switched from aerobic to strictly anaerobic conditions. Which process can still produce ATP?

- A)** fatty-acid β -oxidation feeding the electron transport chain
- B)** glycolysis
- C)** proton flow through ATP synthase
- D)** the Krebs cycle
- E)** oxidative phosphorylation

Answer: B

Glycolysis requires no oxygen and still delivers a net 2 ATP per glucose, so it continues under strictly anaerobic conditions. With O_2 absent the electron transport chain stalls, so oxidative phosphorylation and proton flow through ATP synthase stop, NAD^+ runs short, fermentation must regenerate it, and the Krebs cycle and fatty-acid β -oxidation grind to a halt.

Section C - Chemistry (Q33-47)

33. What is the oxidation number of chromium in the dichromate ion, $Cr_2O_7^{2-}$?

- A)** +3 **B)** +5 **C)** +6 **D)** +7 **E)** +12

Answer: C

Given: the dichromate ion $Cr_2O_7^{2-}$; each oxygen has oxidation number -2 ; the sum of oxidation numbers equals the ion charge.

Relationship: $2 \times (Cr) + 7 \times (O) = -2$, with x the oxidation number of one chromium.

Substitute: $2x + 7 \times (-2) = -2$

$$2x - 14 = -2$$

$$2x = -2 + 14 = +12$$

$$x = +12 \div 2 = +6$$

Result: chromium is +6 in dichromate.

Classic slip: stopping at $2x = +12$ and reporting +12 forgets to divide by the two chromium atoms; +7 is the state of manganese in permanganate, a different ion.

34. In the reaction $Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$, the reducing agent is:

- A)** Zn **B)** H^+ **C)** Cl^- **D)** H_2 **E)** Zn^{2+}

Answer: A

Givens: $Zn + 2HCl \rightarrow ZnCl_2 + H_2$; a reducing agent is the species that itself loses electrons (is oxidised).

Deduction 1: zinc goes from oxidation number 0 as the element to +2 in $ZnCl_2$, so zinc loses 2 electrons and is oxidised.

Deduction 2: hydrogen goes from +1 in H^+ to 0 in H_2 , so hydrogen gains electrons and is reduced.

Deduction 3: the electrons zinc loses are exactly the ones H^+ gains, so zinc causes the reduction.

Conclusion: Zn is the reducing agent.

The tempting choice H^+ fails because it gains electrons: it is the oxidising agent, and chloride ions are unchanged spectators.

35. Which molecule has a net dipole moment (is polar overall)?

- A) CO₂ B) BF₃ C) CCl₄ D) CH₃Cl E) CH₄

Answer: D

CH₃Cl is tetrahedral yet asymmetric: its single C-Cl bond dipole is not balanced by the three C-H dipoles, so a net dipole remains. CO₂ (linear), BF₃ (trigonal planar), CCl₄ and CH₄ (both tetrahedral) are all symmetric, and their individual bond dipoles cancel to zero overall.

36. 0.001 M hydrochloric acid (pH 3) is diluted tenfold with water. The pH of the new solution is:

- A) 2 B) 3 (unchanged) C) 4 D) 5 E) 7

Answer: C

Given: 0.001 M HCl, a strong acid, so $[H^+] = 0.001 \text{ M} = 10^{-3} \text{ M}$ (pH 3); then diluted tenfold with water. Relationship: $\text{pH} = -\log_{10}[H^+]$; a tenfold dilution divides $[H^+]$ by 10 and raises pH by exactly 1 unit.

Substitute: $[H^+]$ after dilution $= 10^{-3} \div 10 = 10^{-4} \text{ M}$

$$\text{pH} = -\log_{10}(10^{-4}) = 4$$

Result: pH 4.

Classic slip: reading dilution as making the acid stronger gives pH 2; dilution always weakens an acid and raises pH, and pH 7 would need dilution so extreme that water's own ionisation takes over.

37. What is the pH of a 0.01 M solution of NaOH at 25 °C? ($K_w = 1.0 \times 10^{-14}$)

- A) 2 B) 10 C) 11 D) 12 E) 13

Answer: D

Given: 0.01 M NaOH at 25 °C; $K_w = 1.0 \times 10^{-14}$; NaOH is a strong base and dissociates completely.

Relationship: $[OH^-] = 10^{-2} \text{ M}$; $\text{pOH} = -\log_{10}[OH^-]$; $\text{pH} + \text{pOH} = 14$ at 25 °C

Substitute: $\text{pOH} = -\log_{10}(10^{-2}) = 2$

$$\text{pH} = 14 - 2 = 12$$

Result: pH 12.

Classic slip: reporting 2 mistakes pOH for pH; a value of 13 would need a ten times more concentrated 0.1 M hydroxide solution.

38. 3 mol of H₂ are mixed with 2 mol of O₂ and sparked: $2H_2 + O_2 \rightarrow 2H_2O$. What is the maximum amount of water formed?

- A) 1.5 mol B) 2 mol C) 3 mol D) 4 mol E) 6 mol

Answer: C

Given: 3 mol H₂ and 2 mol O₂; equation $2H_2 + O_2 \rightarrow 2H_2O$.

Relationship: 2 mol H₂ consume 1 mol O₂ and form 2 mol H₂O; the reactant that runs out first (the limiting one) fixes the yield.

Substitute: O₂ needed to burn all the H₂ $= 3 \times (1 \div 2) = 1.5 \text{ mol O}_2$

1.5 mol needed < 2 mol available, so O₂ is in excess and H₂ is limiting.

Water formed $= 3 \text{ mol H}_2 \times (2 \text{ mol H}_2\text{O} \div 2 \text{ mol H}_2) = 3 \text{ mol H}_2\text{O}$

Result: 3 mol of water.

Classic slip: scaling from the 2 mol of O₂ gives 4 mol of water, using a reactant that is never fully consumed because the hydrogen runs out first.

39. 20 mL of a 1.5 M HCl solution is diluted with water to a final volume of 150 mL. The concentration of the diluted solution is:

- A)** 0.10 M **B)** 0.15 M **C)** 0.20 M **D)** 0.25 M **E)** 0.30 M

Answer: C

Given: 20 mL of 1.5 M HCl; final volume after dilution 150 mL.

Relationship: dilution conserves moles of solute; $C_1V_1 = C_2V_2$.

Substitute: moles of HCl = $0.020 \text{ L} \times 1.5 \text{ mol/L} = 0.030 \text{ mol}$

$C_2 = 0.030 \text{ mol} \div 0.150 \text{ L} = 0.20 \text{ mol/L}$

Result: 0.20 M.

Classic slip: reading 20 mL to 150 mL as a tenfold dilution gives 0.15 M; the true dilution factor is $150 \div 20 = 7.5$.

40. How many molecules are present in 0.50 mol of CO_2 ? ($N_a = 6.02 \times 10^{23} \text{ mol}^{-1}$)

- A)** 1.20×10^{24} **B)** 3.01×10^{22} **C)** 3.01×10^{23} **D)** 6.02×10^{23} **E)** 9.03×10^{23}

Answer: C

Given: $n = 0.50 \text{ mol}$ of CO_2 ; $N_a = 6.02 \times 10^{23} \text{ mol}^{-1}$

Relationship: number of molecules = moles $\times$ Avogadro's constant.

Substitute: $N = 0.50 \times 6.02 \times 10^{23} = 3.01 \times 10^{23}$ molecules

Result: 3.01×10^{23} molecules.

Classic slip: multiplying by 3 again to get 9.03×10^{23} counts the atoms (three per CO_2 molecule), not the molecules the question asks for.

41. A sealed elastic balloon at constant pressure holds air at 27°C . It is warmed to 327°C . Its volume:

- A)** increases about 1.1-fold
B) stays the same
C) increases about 12-fold
D) halves
E) doubles

Answer: E

Given: constant pressure; initial temperature 27°C , final temperature 327°C .

Relationship: Charles's law, V proportional to T at constant pressure, with T in kelvin ($K = ^\circ\text{C} + 273$).

Substitute: $T_1 = 27 + 273 = 300 \text{ K}$

Substitute: $T_2 = 327 + 273 = 600 \text{ K}$

Volume ratio = $T_2 \div T_1 = 600 \div 300 = 2$

Result: the volume doubles.

Classic slip: dividing the Celsius values ($327 \div 27 \approx 12$) predicts a spurious twelve-fold increase; gas laws never use Celsius.

42. Which atom has the largest radius?

- A)** Mg **B)** K **C)** Na **D)** Cl **E)** Al

Answer: B

Atomic radius increases down a group and decreases across a period. Potassium lies one full period below sodium, magnesium, aluminium and chlorine, at the left edge of its own period, so it is the largest atom listed; within the third period the ranking runs $\text{Na} > \text{Mg} > \text{Al} > \text{Cl}$.

43. A white crystalline solid has a high melting point, shatters when struck, and conducts electricity only when molten or dissolved. It is most likely:

- A)** a network covalent solid such as quartz
- B)** a covalent molecular solid such as wax
- C)** a metal such as copper
- D)** graphite
- E)** an ionic compound such as NaCl

Answer: E

High melting point, brittleness under impact, and conduction only when molten or dissolved are the classic signature of an ionic solid: a rigid lattice of ions that cannot carry charge until free to move, and that shatters because slipped layers bring like charges together and repel. Metals conduct as solids, molecular solids such as wax melt far too easily, and graphite conducts even as a solid through delocalised electrons.

44. Ethanoic acid reacts with ethanol in the presence of a few drops of concentrated H_2SO_4 . The organic product is:

- A)** ethyl ethanoate
- B)** ethanal
- C)** ethanoic anhydride
- D)** sodium ethanoate
- E)** ethanol

Answer: A

An acid-catalysed reaction of a carboxylic acid with an alcohol is esterification: ethanoic acid + ethanol $\rightarrow$ ethyl ethanoate + water, with the concentrated H_2SO_4 acting as catalyst. Sodium ethanoate would need a base such as NaOH, ethanal is a mild oxidation product of ethanol, and the anhydride would require two acid molecules to condense rather than an acid with an alcohol.

45. Which single observation distinguishes ethene from ethane?

- A)** Both burn in air with a luminous flame.
- B)** Ethane is a gas; ethene is a liquid at room temperature.
- C)** Ethene dissolves freely in water.
- D)** Ethane contains a carbon-carbon double bond.
- E)** Ethene decolourises bromine water; ethane does not (under these conditions).

Answer: E

Ethene's carbon-carbon double bond adds bromine across itself immediately, decolourising bromine water, whereas saturated ethane reacts with bromine only slowly under ultraviolet light, giving one clean visual test. Both hydrocarbons are gases at room temperature, neither dissolves freely in water, and the double bond belongs to ethene, not ethane.

46. For the exothermic equilibrium $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$, raising the temperature at constant pressure:
- A) increases both the rate and the equilibrium yield
 - B) decreases the rate but increases the yield
 - C) affects neither rate nor yield
 - D) increases the rate but decreases the equilibrium yield
 - E) increases the rate and leaves the yield unchanged

Answer: D

Raising the temperature always increases the rate, because more molecules exceed the activation energy, so any claim of a slower reaction fails. For this exothermic equilibrium, Le Chatelier's principle shifts the position of equilibrium in the endothermic (reverse) direction at higher temperature, so the equilibrium yield of SO_3 falls. The correct combination is therefore a faster reaction with a lower yield.

47. A solution contains 0.10 mol of glucose dissolved in 500 g of water. Its molality is:
- A) 0.05 mol/kg B) 0.10 mol/kg C) 0.20 mol/kg D) 0.50 mol/kg E) 2.0 mol/kg

Answer: C

Given: 0.10 mol of glucose dissolved in 500 g of water.

Relationship: molality = moles of solute $\div$ kilograms of solvent (solvent mass only, never the whole solution).

Substitute: solvent mass = 500 g = 0.500 kg

molality = 0.10 mol $\div$ 0.500 kg = 0.20 mol/kg

Result: 0.20 mol/kg.

Classic slip: using a solution-based mass or rescaling to a full kilogram ignores that the definition is pinned to the solvent alone.

Section D - Mathematics (Q48-54)

48. Solve the inequality $5 - 3x \geq 14$.
- A) $x < -3$ B) $x \leq -3$ C) $x \leq 3$ D) $x \geq 3$ E) $x \geq -3$

Answer: B

Given: $5 - 3x \geq 14$.

Relationship: isolate x ; dividing an inequality by a negative number reverses the direction of the sign.

Step 1: subtract 5 from both sides: $-3x \geq 14 - 5 = 9$.

Step 2: divide by -3 and flip the sign: $x \leq 9 \div (-3) = -3$

Result: $x \leq -3$.

Classic slip: dividing by -3 without flipping gives $x \geq -3$; the boundary itself is included, since $5 - 3 \times (-3) = 14$ satisfies the original exactly.

49. What is the domain of $f(x) = \sqrt{(4 - x) / (x - 2)}$?

- A) $x \leq 4$ with $x \neq 2$
- B) $x \leq 4$ with $x \neq 4$
- C) $x \leq 4$
- D) $x < 4$
- E) all real numbers

Answer: A

Given: $f(x) = \sqrt{(4 - x) / (x - 2)}$; a square root requires a non-negative input, and a denominator may not be zero.

Condition 1: $4 - x \geq 0$, so $x \leq 4$.

Condition 2: $x - 2 \neq 0$, so $x \neq 2$.

Combine: $x \leq 4$ and $x \neq 2$.

Result: the domain is $x \leq 4$ with $x \neq 2$.

Classic slip: applying only the square-root condition keeps $x = 2$, where the function divides by zero; excluding $x = 4$ is groundless, since $x = 4$ simply makes $f(x) = 0$.

50. What is the value of $\log_3 81$?

- A) 3 B) 4 C) 9 D) 27 E) 81

Answer: B

Given: $\log_3 81$.

Relationship: $\log_3 81 = x$ means $3^x = 81$

Build the powers: $3^1 = 3$; $3^2 = 9$; $3^3 = 27$; $3^4 = 81$

Four factors of 3 are needed, so $x = 4$

Result: 4.

Classic slip: answering 27 answers the reversed question (what 3^3 equals) instead of the exponent that produces 81.

51. Three fair coins are tossed. What is the probability of getting at least one head?

- A) $1/2$ B) $1/8$ C) $2/3$ D) $3/8$ E) $7/8$

Answer: E

Given: three fair coins tossed once each; $P(\text{head}) = P(\text{tail}) = 1/2$, tosses independent.

Relationship: $P(\text{at least one head}) = 1 - P(\text{no heads})$, and independent probabilities multiply.

Substitute: $P(\text{no heads}) = 1/2 \times 1/2 \times 1/2 = 1/8$

$P(\text{at least one head}) = 1 - 1/8 = 7/8$

Result: $7/8$.

Classic slip: $3/8$ is the probability of exactly one head (three of the eight equally likely outcomes), a different event from at least one head.

52. A straight ramp 10 m long makes a 60° angle with the horizontal ground. How far along the ground does the ramp span (horizontal distance between its two ends)? ($\cos 60^\circ = 0.5$, $\sin 60^\circ = 0.87$)

- A)** 2.5 m **B)** 5 m **C)** 6.7 m **D)** 8.7 m **E)** 10 m

Answer: B

Given: ramp length 10 m; angle with the horizontal 60° ; $\cos 60^\circ = 0.5$, $\sin 60^\circ = 0.87$.

Relationship: the ground span is the side adjacent to the angle, so $\text{span} = \text{hypotenuse} \times \cos(\text{angle})$.

Substitute: $\text{span} = 10 \text{ m} \times \cos 60^\circ = 10 \times 0.5 = 5 \text{ m}$

Result: 5 m.

Classic slip: using sine instead ($10 \times 0.87 = 8.7 \text{ m}$) computes the ramp's vertical height, not the horizontal distance along the ground.

53. If $f(x) = 2x - 3$ and $g(x) = x^2$, what is $f(g(2))$?

- A)** 1 **B)** 2 **C)** 4 **D)** 5 **E)** 8

Answer: D

Given: $f(x) = 2x - 3$ and $g(x) = x^2$; required $f(g(2))$

Relationship: $f(g(2))$ applies g first, then feeds its output into f (work inside out).

Step 1: $g(2) = 2^2 = 4$

Step 2: $f(4) = 2 \times 4 - 3 = 8 - 3 = 5$

Result: 5.

Classic slip: reporting 8 keeps only the doubling and drops the -3 ; running the composition backwards gives $g(f(2)) = (2 \times 2 - 3)^2 = 1$, a different question.

54. Simplify $(2^5 \times 2^3) \div 2^4$.

- A)** 4 **B)** 8 **C)** 16 **D)** 32 **E)** 64

Answer: C

Given: $(2^5 \times 2^3) \div 2^4$.

Relationship: with equal bases, multiplying adds exponents and dividing subtracts them.

Step 1: $2^5 \times 2^3 = 2^{5+3} = 2^8$

Step 2: $2^8 \div 2^4 = 2^{8-4} = 2^4$

Step 3: $2^4 = 16$

Result: 16.

Classic slip: subtracting in the wrong order (2^{4-8}) or multiplying the exponents (2^{15}) breaks the index laws; the correct chain is add, then subtract.

Section E - Physics (Q55-60)

55. A 6 V battery drives a current through a $2\ \Omega$ resistor connected in series with a parallel pair made of a $6\ \Omega$ and a $3\ \Omega$ resistor. What current flows from the battery?

- A)** 0.5 A **B)** 1.0 A **C)** 1.5 A **D)** 3.0 A **E)** 6.0 A

Answer: C

Given: battery 6 V; a $2\ \Omega$ resistor in series with a parallel pair of $6\ \Omega$ and $3\ \Omega$.

Relationship: parallel pair: $1/R = 1/R_1 + 1/R_2$; series resistances add; current $I = V \div R$.

Step 1: $1/R = 1/6 + 1/3 = 1/6 + 2/6 = 3/6 = 1/2$, so the pair is $2\ \Omega$

Step 2: total resistance = $2\ \Omega + 2\ \Omega = 4\ \Omega$

Step 3: $I = 6\ \text{V} \div 4\ \Omega = 1.5\ \text{A}$

Result: 1.5 A.

Classic slip: dividing 6 V by the parallel pair alone ($2\ \Omega$) forgets the series resistor; adding all three as a series chain ($11\ \Omega$) ignores the parallel connection.

56. A kettle draws 4.6 A from a 230 V supply. Its power consumption is closest to:

- A)** 50 W **B)** 230 W **C)** 1060 W **D)** 2300 W **E)** 4600 W

Answer: C

Given: $V = 230\ \text{V}$ and $I = 4.6\ \text{A}$

Relationship: electrical power $P = V \times I$.

Substitute: $P = 230 \times 4.6 = 1058\ \text{W}$

$1058\ \text{W} \approx 1060\ \text{W}$.

Result: about 1060 W.

Classic slip: dividing instead of multiplying ($230 \div 4.6 = 50$) produces the 50 W value.

57. Equal masses of aluminium (specific heat $900\ \text{J}/(\text{kg}\cdot\text{K})$) and copper (specific heat $385\ \text{J}/(\text{kg}\cdot\text{K})$) receive the same quantity of heat. Which statement is correct?

- A)** Copper's temperature rises about 2.3 times more than aluminium's.
B) Both rise by the same amount.
C) Copper's temperature does not change.
D) It cannot be determined without knowing the exact heat supplied.
E) Aluminium's temperature rises more than copper's.

Answer: A

Given: equal masses of aluminium and copper; equal heat Q supplied; $c(\text{Al}) = 900\ \text{J}/(\text{kg}\cdot\text{K})$, $c(\text{Cu}) = 385\ \text{J}/(\text{kg}\cdot\text{K})$.

Relationship: $Q = m c \Delta T$, so $\Delta T = Q \div (m c)$; with Q and m equal, ΔT is proportional to $1/c$

Substitute: $\Delta T(\text{Cu}) \div \Delta T(\text{Al}) = c(\text{Al}) \div c(\text{Cu}) = 900 \div 385 \approx 2.3$

So copper, with the smaller specific heat, warms about 2.3 times more.

Result: copper's temperature rises about 2.3 times more than aluminium's.

Classic slip: expecting the larger specific heat to give the larger rise reverses the inverse relationship; the actual value of Q cancels in the ratio, so it is not needed.

58. A car travelling at 20 m/s brakes uniformly and comes to rest in 4.0 s. The magnitude of its deceleration is:

- A)** 0.2 m/s² **B)** 4 m/s² **C)** 5 m/s² **D)** 16 m/s² **E)** 20 m/s²

Answer: C

Given: initial speed 20 m/s, final speed 0, stopping time 4.0 s.

Relationship: for uniform braking, $a = \Delta v \div t = (\text{final} - \text{initial speed}) \div t$; the magnitude is the positive size of this.

Substitute: $a = (20 - 0) \div 4.0 = 20 \div 4.0 = 5 \text{ m/s}^2$

Result: 5 m/s².

Classic slip: dividing the other way ($4.0 \div 20 = 0.2$) inverts the definition of acceleration.

59. A velocity-time graph shows: a linear rise from 0 to 4 m/s during the first 4 s, a constant 4 m/s for the next 4 s, then a linear fall back to 0 over the final 4 s. What total distance is covered?

- A)** 16 m **B)** 24 m **C)** 32 m **D)** 40 m **E)** 48 m

Answer: C

Given: speed rises 0 to 4 m/s during 0 to 4 s; holds 4 m/s from 4 to 8 s; falls to 0 during 8 to 12 s.

Relationship: distance = area under the velocity-time graph.

Step 1, acceleration triangle: $\frac{1}{2} \times 4 \text{ s} \times 4 \text{ m/s} = 8 \text{ m}$

Step 2, constant-speed rectangle: $4 \text{ s} \times 4 \text{ m/s} = 16 \text{ m}$

Step 3, braking triangle: $\frac{1}{2} \times 4 \text{ s} \times 4 \text{ m/s} = 8 \text{ m}$

Total distance = $8 + 16 + 8 = 32 \text{ m}$

Result: 32 m.

Classic slip: counting only the flat rectangle (16 m) drops both triangles; the trapezium formula $\frac{1}{2} \times (12 + 4) \times 4 = 32 \text{ m}$ confirms the total.

60. A block floats stationary in water. Which statement must be true?

- A)** The buoyant force is zero.
B) The density of the block is greater than that of water.
C) The mass of displaced water equals the volume of the block.
D) The weight of the displaced water equals the weight of the block.
E) The block displaces a volume of water equal to its own volume.

Answer: D

Givens: the block floats stationary in water; stationary means zero net force; Archimedes' principle says the buoyant force equals the weight of the displaced fluid.

Deduction 1: zero net force means the upward buoyant force equals the block's downward weight.

Deduction 2: that buoyant force equals the weight of the displaced water.

Deduction 3: so the weight of the displaced water equals the weight of the block.

Conclusion: the displaced water weighs exactly as much as the block.

The equal-volume claim would need the block's density to match water's and the block fully submerged; a floater is less dense than water and displaces less volume than its own.